PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 2 • Chapter OL-T27 Classified

Cross-session • chapter-organised candidate

24 classified items • 103 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 2 • Unit 2 • Chapter OL-T27 • 24 items • 103 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Simultaneous Equations

Unit 2 • 103 marks • 24 item(s)

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

22 Solve the simultaneous equations

$$x^2 + 3y + y^2 = 7$$
$$y = x + 2$$

Show clear algebraic working.

....

(Total for Question 22 is 5 marks)

WORKING SPACE

8 Solve the simultaneous equations

$$3x + 5y = 8$$
$$4x + y = -3.5$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 8 is 3 marks)

WORKING SPACE

Question 16

4MA1-2H Q10

MODULAR UNIT 2
3 marks

10 Solve the simultaneous equations

$$6x + 4y = 1$$
$$3x + 5y = 8$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

WORKING SPACE

Question 25

4MA1-2H Q22

MODULAR UNIT 2
5 marks

22 The straight line **L** has equation $x + y = 5$

The curve **C** has equation $2x^2 + 3y^2 = 210$

Find the coordinates of the points where **L** and **C** intersect.
Show clear algebraic working.

(.......................... , ...) (.......................... , ...)

WORKING SPACE

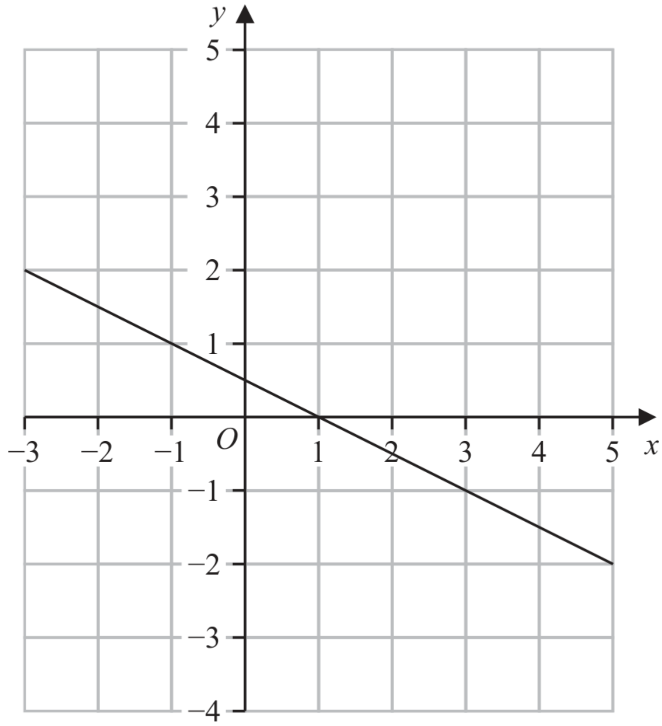
Question 29

4MA1-2H Q14 whole

MODULAR UNIT 2

3 marks

14 Here is the graph of the equation $2y + x = 1$ drawn on a grid.



By drawing another straight line on the grid, solve the simultaneous equations

$$y - x - 2 = 0$$
$$2y + x = 1$$

$x =$ ....

$y =$ ....

Question 34
4MA1-2H Q21 whole

MODULAR UNIT 2
5 marks

21 The line with equation $x + 2y = 5$ intersects the curve with equation $x^2 + 3y^2 = 13$ at the points A and B

Find the coordinates of A and the coordinates of B
Show clear algebraic working.

(.............................. , ...)

(.............................. , ...)

WORKING SPACE

Question 09

4MA1-1H Q9

MODULAR UNIT 2

3 marks

9 Solve the simultaneous equations

$$2x + 9y = 14.5$$
$$7x + 3y = 8$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

WORKING SPACE

Question 28
4MA1-2H Q21

MODULAR UNIT 2
5 marks

21 Solve the simultaneous equations

$$2x^2 + 3y^2 = 11$$
$$x = 3y - 1$$

Show clear algebraic working.

....

WORKING SPACE

Question 34

4MA1-2H Q22 M01

MODULAR UNIT 2
5 marks

22 Solve the simultaneous equations

$$x^2 + y^2 + y = 3$$
$$x + 2 = y$$

Show clear algebraic working.

....

WORKING SPACE

Question 24

4MA1-2H Q10 Q10

MODULAR UNIT 2
3 marks

10 Solve the simultaneous equations

$$7x + 3y = 3$$
$$3x - y = 7$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

WORKING SPACE

Question 12

4MA1-1H Q19

MODULAR UNIT 2

5 marks

19 Solve the simultaneous equations

$$3x^2 + y^2 - xy = 5$$
$$y = 2x - 3$$

Show clear algebraic working.

....

WORKING SPACE

Question 24

4MA1-2H Q10

MODULAR UNIT 2

3 marks

10 Solve the simultaneous equations

$$3x + 5y = 3.1$$
$$6x + 3y = 3.75$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

WORKING SPACE

22 The line with equation $y = x + 2$ intersects the curve with equation $x^2 + y^2 - 2y = 24$ at the points A and B.

Find the coordinates of A and B.
Show clear algebraic working.

(.............................. , ...)

(.............................. , ...)

YOUR WORKING

January 2021 | 4MA1-1H | Question 19 | 5 marks

Chapter OL-T27: Simultaneous Equations

19 Solve the simultaneous equations

$$x^2 - 9y - x = 2y^2 - 12$$
$$x + 2y - 1 = 0$$

Show clear algebraic working.

....

YOUR WORKING

8 Solve the simultaneous equations

$$5a + 2c = 10$$
$$2a - 4c = 7$$

Show clear algebraic working.

$a =$ ....

$c =$ ....

YOUR WORKING

June 2021 | 4MA1-2H | Question 19 | 5 marks

Chapter OL-T27: Simultaneous Equations

19 Solve the simultaneous equations

$$y = 3 - 2x$$
$$x^2 + y^2 = 18$$

Show clear algebraic working.

....

YOUR WORKING

6 Alison buys 5 apples and 3 pears for a total cost of \$1.96
Greg buys 3 apples and 2 pears for a total cost of \$1.22

Michael buys 10 apples and 10 pears.

Work out how much Michael pays for his 10 apples and 10 pears.
Show your working clearly.

\$..

YOUR WORKING

19 The straight line **L** has equation $x - y = 3$
The curve **C** has equation $3x^2 - y^2 + xy = 9$

L and **C** intersect at the points P and Q.

Find the coordinates of the midpoint of PQ.
Show clear algebraic working.

(............................ , ...)

YOUR WORKING

12 Solve the simultaneous equations

$$7x - 2y = 34$$
$$3x + 5y = -3$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 12 is 4 marks)

YOUR WORKING

May-Nov 2020 | 2H | Question 16 | 5 marks

Chapter OL-T27: Simultaneous Equations

16 Solve the simultaneous equations

$$3xy - y^2 = 8$$
$$x - 2y = 1$$

Show clear algebraic working.

....

(Total for Question 16 is 5 marks)

YOUR WORKING

January 2023 | 2H | Question 7 | 3 marks

Chapter OL-T27: Simultaneous Equations

7 Solve the simultaneous equations

$$5x + 4y = -2$$
$$2x - y = 4.4$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 7 is 3 marks)

YOUR WORKING

January 2023 | 2H | Question 20 | 5 marks

Chapter OL-T27: Simultaneous Equations

20 Solve the simultaneous equations

$$y = 7 - 2x$$
$$x^2 + y^2 = 34$$

Show clear algebraic working.

....

(Total for Question 20 is 5 marks)

YOUR WORKING

November 2025 | 1H | Question 28 | 6 marks

Chapter OL-T27: Simultaneous Equations

28 The curve **C** has equation $x^2 + y^2 = d - 11x$ where d is an integer.
The line **L** has equation $y = 3x + e$ where e is an integer.

C and **L** intersect at the point A and at the point B

The coordinates of A are (0.2, 2.6)

Work out the coordinates of B
Show your working clearly.

YOUR WORKING

November 2025 | 1H | Question 28 | 6 marks

Chapter OL-T27: Simultaneous Equations

(.......................... , ...)

(Total for Question 28 is 6 marks)

YOUR WORKING

November 2025 | 2H | Question 2 | 3 marks

Chapter OL-T27: Simultaneous Equations

2 Solve the simultaneous equations

$$5x + y = 11$$
$$3x - y = 9$$

Show clear algebraic working.

$x =$ ....

$y =$ ....

(Total for Question 2 is 3 marks)

YOUR WORKING